

Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

PROGRAM:

```
#include <reg51.h>

void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
    unsigned long sum = 0;

    for(i = 0; i < 5; i++)
    {
        sum += array[i];
    }

    P0 = (unsigned char)(sum & 0xFF);
    P1 = (unsigned char)((sum >> 8) & 0xFF);
    P2 = (unsigned char)((sum >> 16) & 0xFF);
    P3 = (unsigned char)((sum >> 24) & 0xFF);

    while(1);
}
```

CUUsers\koshal\OneDrive\Documents\Embedded System MANO\EXP 13\exp13 - uVision

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Registers

Register	Value
R0	0x00
R1	0x00
R2	0x00
R3	0x00
R4	0x00
R5	0x00
R6	0x00
R7	0x00
R8	0x00
R9	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x00
R17	0x00
R18	0x00
R19	0x00
R20	0x00
R21	0x00
R22	0x00
R23	0x00
R24	0x00
R25	0x00
R26	0x00
R27	0x00
R28	0x00
R29	0x00
R30	0x00

Disassembly

```
1: P2 = (unsigned char)((sum >> 16) & 0xFF);
2:
3: C:\0x094A AF55 MOV R7, #0x15
4: C:\0x094C AE14 MOV R6, #0x14
5: C:\0x094E AD13 MOV R5, #0x13
6: C:\0x0970 AC12 MOV R4, #0x12
7: C:\0x0972 7B10 MOV R0, #0x10
8:
9:
10:
11:
12:
13:
14: P0 = (unsigned char)(sum & 0xFF);
15: P1 = (unsigned char)((sum >> 8) & 0xFF);
16: P2 = (unsigned char)((sum >> 16) & 0xFF);
17: P3 = (unsigned char)((sum >> 24) & 0xFF);
18:
19: while(1);
20: }
```

STARTUP.AS1

```
1: #include <reg51.h>
2:
3: void main(void)
4: {
5:     unsigned int i;
6:     unsigned int array[] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7:     unsigned long sum = 0;
8:     for(i = 0; i < 5; i++)
9:     {
10:         sum += array[i];
11:     }
12:
13:     P0 = (unsigned char)(sum & 0xFF);
14:     P1 = (unsigned char)((sum >> 8) & 0xFF);
15:     P2 = (unsigned char)((sum >> 16) & 0xFF);
16:     P3 = (unsigned char)((sum >> 24) & 0xFF);
17:
18:     while(1);
19: }
```

Parallel Port 0

Port 0	2 Bits	0
P0	0x00	0x00
Prev	0x00	0x00

Parallel Port 1

Port 1	2 Bits	0
P1	0xAB	0xAB
Prev	0xAB	0xAB

Parallel Port 2

Port 2	2 Bits	0
P2	0x01	0x01
Prev	0x01	0x01

Parallel Port 3

Port 3	2 Bits	0
P3	0x00	0x00
Prev	0x00	0x00

Command

Running with Code Size Limit: 2K

Load "C:\Users\koshal\OneDrive\Documents\Embedded System MANO\Objects\EXP 13"

Watch 1

Name	Value	Type
<Enter expression>	cannot evaluate	uchar
<Enter expression>	0x00	int
<Enter expression>	array[4]	uint
<Enter expression>	0xABCD	uint

ASH ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE DEFINE DIR Display

Call Stack - Locals Watch 1 Watch 2 Memory 1

Simulation

11:55:33.4366077 sec L16 C46 GAP NAME SCRI CUD 0/24

33°C Mostly cloudy

Search

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